

By 2010s, the studies suggested that neural stimulation could restore functional connectivity and associated behaviours through modulation of molecular mechanisms of synaptic efficacy. In 2011 O'Doherty demonstrated a BCI that enabled a monkey to brain control the position of an avatar arm by receiving sensory feedback through direct intracortical stimulation in the sensory cortex.

Recently, in 2019, a BCI showcased usage of deep learning to synthesize speech to help patients suffering from speech impairment due to neurological disorders. In 2020, Elon Musk's Neuralink was successfully implanted in a pig and the University of California demonstrated a computing system

to translate brainwaves into sentences with the word error rates as low as 3%. In 2021 Musk announced that a monkey could play video games using a BCI device.

Electromyography is the recording of the electrical activity of muscle tissue, or its representation as a visual display or audible signal, using electrodes attached to the skin or inserted into the muscle. BCIs that predict electromyographic activity are being developed to restore mobility in paralysed limbs.

The U.S. Army Research Laboratory has recently developed a mobile BCI device that measures steady state visually evoked potentials (SSVEPs), which are signals that are natural

responses to visual stimulation at specific frequencies. Mind-controlled wheelchairs and home automation applications have been demonstrated being operated by people with limited or no motor control.

TYPES OF BCIs

Invasive BCIs

Invasive BCIs use electrodes implanted inside the skull to sense the electrical activity produced by the brain cells. As these lie in the grey matter, they produce the highest quality signals. However, they are prone to scar-tissue build-up due to body reaction to the presence of foreign objects causing them to become weaker or even non-existent with the passage of time. In some cases, the body may not accept the electrodes implanted into brain causing a medical condition.

Generally, the invasive BCIs are used for repairing damaged sight and providing new functionality for people with paralysis. For example, a person using a BrainGate chip-implant could control a robotic arm as well as a computer cursor, lights and TV just by thinking about the activity.

Partially invasive BCIs

Partially invasive BCI devices are implanted inside the skull but outside the grey matter. They produce signals with resolution higher than non-invasive BCIs and have a lower risk of forming scar-tissue in the brain. It has also been possible for these to be inserted via the vascular system, eliminating the need for open brain surgery. In another technology, called electrocorticalography (ECoG), the electrodes are placed above the cortex, beneath the dura mater (outermost membrane enveloping the brain and spinal cord).

Partially invasive technology has lower complexity, lower clinical risk and superior long-term stability. They provide higher spatial resolution, better signal-to-noise ratio,

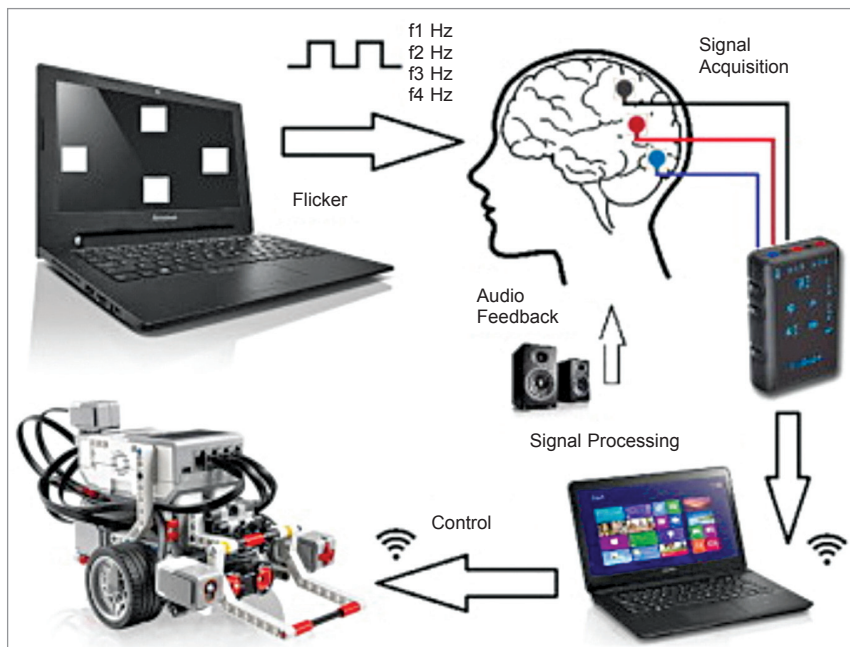


Fig. 2: General structure of SSVEP based BCI (Source: www.sciencedirect.com)

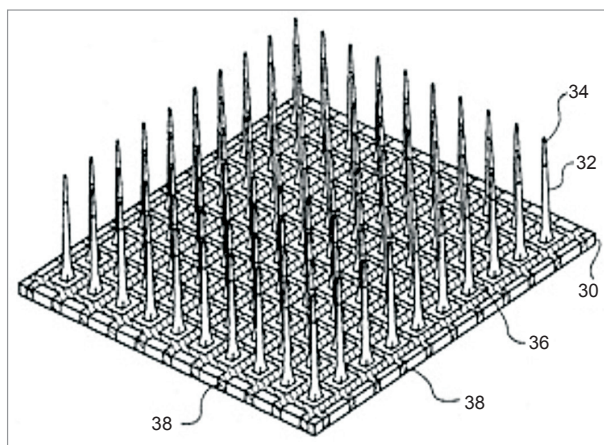


Fig. 3: An invasive BCI (Source: <http://learn.neurotechedu.com/introtobci>)

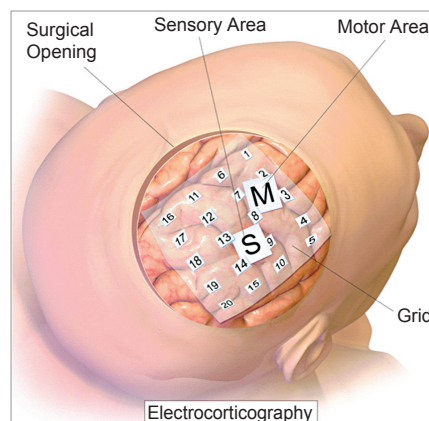


Fig. 4: A partially invasive BCI (Source: <http://learn.neurotechedu.com/introtobci>)